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GP2 Research Report

All primary empirical materials referenced throughout this Research Report are included in the appendices for transparency and traceability. Full interview transcripts are provided in Appendix A. Whenever an interview is cited in the main text, the corresponding transcript can be found in Appendix A under the relevant track designation.

Appendix B contains the complete VCH Lab Case Study Insight Analysis that informed the problem diagnosis. Appendix C provides a structured overview of desk research sources and their relevance to conceptual and methodological decisions. Appendix D summarises the methodological development process, and Appendix E contains formal evidence of passing the approved Graduation Project Plan.

These appendices are included to ensure that all analytical claims remain verifiable and that the reasoning process underlying the study is fully transparent.

This study progressed from methodological design into data collection within the Value Chain Hackers Lab at Windesheim University of Applied Sciences. The lab functions as an applied research environment in which student teams collaborate with real organisations on challenges related to transparency, sustainability, and system resilience. Because projects operate within fixed academic timelines and are shaped by practical constraints such as variation in research experience and stakeholder availability, the research was conducted in an uncontrolled real-world setting. These contextual conditions informed both the sequencing and execution of the methodology.

The Graduation Project Plan defined an initial qualitative research direction supported by desk research and field based inquiry. During GP2, this direction was strengthened rather than replaced. Extended desk research substantiated the legitimacy of the research problem and identified theoretically grounded constructs related to documentation, traceability, continuity, and reproducibility. These constructs were translated into examinable dimensions through the development of an operationalisation framework, which clarified how abstract concepts would be empirically assessed and ensured alignment between research questions, data sources, and analytical logic.

Following this conceptual grounding, the study entered the empirical phase using complementary sources of evidence. Systematic document analysis examined how documentation practices materialise within completed VCH Lab project outputs, while structured interviews with the lab supervisor and former team leaders provided insight into how these practices are enacted during project execution. The interview guide was constructed after operationalisation by translating validated constructs into thematic blocks, core questions, and targeted probes, ensuring methodological coherence and instrument validity.

The sequencing of desk research, operationalisation, document analysis, and interviews reflects an intentionally structured data collection strategy. Conceptual grounding preceded empirical inquiry, empirical findings informed interpretive focus, and multiple sources were used to examine the same constructs from different perspectives. Together, these elements establish a transparent and methodologically defensible foundation for answering the research question within the defined institutional scope.

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Below is the revised version.

1. Data Collection

1.1 Research Context and Design Execution

This study was conducted within the Value Chain Hackers Lab at Windesheim University of Applied Sciences, an applied research environment where student teams collaborate with external organisations on sustainability, transparency, and supply chain resilience challenges. Projects operate within fixed academic timelines and are shaped by variation in research experience, stakeholder availability, evolving scopes, and institutional expectations. The study was therefore conducted in an uncontrolled real world setting in which variables are not manipulated but practices are observed as they naturally occur, consistent with case study research in real environments (Yin, 2018).

The research investigates documentation practices, traceability of reasoning, and continuity across student cohorts. Central is the concept of provenance, defined as the documented lineage of datasets, analytical decisions, transformations, and interpretive steps that enables reconstruction of how conclusions were reached. Provenance functions as an evaluative criterion for continuity rather than an abstract principle. Reproducibility scholarship identifies transparency of analytical pathways and preservation of provenance as central to research credibility (Peng, 2011; Wilkinson et al., 2016). Lifecycle models further emphasise that documentation must extend beyond final reporting to structured preservation and reuse across research phases and cohort handovers (Gooch et al., 2019; Mons et al., 2017). The VCH Lab therefore represents a micro level setting in which recognised continuity vulnerabilities can be empirically examined.

Institutional Benchmarking

Documentation continuity is not a local concern. National funding bodies require formal research data management planning (NWO, n.d.). International governance frameworks emphasise accessibility, reuse, and transparent research design (OECD, 2021; Science Europe, 2021). Universities operationalise FAIR aligned policies (University of Twente, n.d.), while infrastructure providers such as SURF and 4TU.ResearchData support lifecycle preservation through metadata standards and persistent identifiers (SURF, n.d.; 4TU.ResearchData, n.d.). Verification initiatives such as reproducibility checks at TU Delft reinforce expectations of independent verifiability (TU Delft Digital Competence Centre, n.d.). Together, these developments demonstrate that documentation embedded within lifecycle processes is a recognised structural requirement rather than an optional practice. The IS situation observed within the VCH Lab therefore reflects a broader continuity challenge rather than an isolated operational weakness.

Design Execution and Process Structure

The Graduation Project Plan defined a qualitative design supported by desk research, document analysis, and stakeholder interviews. At the GP1 stage, the design was specified strategically; operationalisation and instrument precision were deferred until theoretical constructs were validated. During GP2, the design was refined rather than replaced. The three core data sources were executed as planned, while conceptual grounding and procedural precision were strengthened. This reflects methodological refinement rather than deviation, consistent with design science reasoning in which conceptual clarity precedes implementation (Hevner et al., 2004; Peffers et al., 2007).

The research proceeded in three phases. First, desk research validated the research problem and identified theoretically supported constructs related to transparency, documentation practices, knowledge continuity, and reproducibility, ensuring analysis through academically grounded constructs rather than anecdotal impressions (Gregor & Hevner, 2013). Second, these constructs were operationalised into observable dimensions aligned with research questions and empirical instruments (Kallio et al., 2016; Patton, 2015). Third, empirical data collection was conducted through systematic document analysis of completed VCH Lab projects and fully structured interviews with relevant stakeholders. Document analysis preceded interviews to preserve empirical independence. To structure where traceability activities should occur across the research lifecycle, the Electronic Discovery Reference Model was used as a sequencing reference. EDRM provides a stage based representation of information identification, processing, review, preservation, and production within complex environments. In this study, it functions solely as a lifecycle mapping aid to determine where documentation, decision capture, preservation, and validation should occur to sustain provenance across research stages. It does not define theoretical constructs or justify methodological choices, thereby preserving theoretical independence while strengthening internal coherence.

No fundamental methodological shifts occurred between GP1 and GP2. An anticipated survey was omitted due to limited participant availability during academic holidays. Rather than administering a survey likely to yield insufficient responses, it was removed to prioritise informational depth over numerical breadth, consistent with qualitative standards emphasising depth in applied contexts (Patton, 2015). The final design maintains continuity with GP1 while

demonstrating deliberate refinement in response to theoretical clarification and practical constraints. All methodological decisions have traceable intellectual origins grounded in field evidence, validated constructs, established models, or methodological literature. The research was conducted in accordance with its intended direction, and refinements were transparent and academically defensible.

1.2 Data Sources and Methods

The data collection strategy followed the staged logic established in Section 1.1. Empirical observation of documentation practices preceded theoretical consolidation to preserve analytical independence. The sequence consisted of identifying the IS situation, validating constructs through desk research, and operationalising those constructs to guide empirical instruments.

As outlined in Section 1.1, project lifecycles were lightly mapped against the Electronic Discovery Reference Model solely as a sequencing aid to assess where documentation, decision capture, preservation, and validation occurred across research stages (EDRM, n.d.).

Three complementary sources of evidence were used: systematic document analysis, fully structured qualitative interviews, and staged desk research. Together they provide triangulated insight across preserved outputs, enacted practice, and theoretical grounding.

Document Analysis

Document analysis examined how documentation practices materialise across complete VCH Lab project lifecycles. Five tracks were analysed:

The Bakery project examined sustainability practices at Bbrood.

The CRM project addressed critical resource management for Gemeente Zwolle and local battery collection agencies, focusing on coordination and vulnerability in municipal waste systems.

The Cacao project explored transparency challenges affecting smallholder cacao farmers in African production regions.

The Textile project investigated sustainability in textile supply chains.

The Organizational project analysed internal research workflow structuring within the lab.

Including varied empirical domains strengthens structural validity by demonstrating that documentation patterns are not confined to a single thematic context.

The analysis assessed documentation completeness, visibility of decision rationales, traceability of analytical steps, version continuity, source transparency, and cross cohort reuse potential. Examining complete project lifecycles aligns with case study methodology emphasising

investigation of real world phenomena within uncontrolled environments while preserving a traceable chain of evidence (Yin, 2018).

These analytical dimensions reflect established concerns in reproducibility and research data lifecycle scholarship. Transparent analytical provenance and recoverable decision pathways are central to research credibility (Peng, 2011). FAIR standards emphasise findability, accessibility, interoperability, and reuse (Wilkinson et al., 2016), while lifecycle models stress structured preservation across planning, execution, analysis, and archiving stages (Gooch et al., 2019; Mons et al., 2017). Governance frameworks further reinforce expectations of accessibility and reuse (OECD, 2021; Science Europe, 2021).

Observed gaps were interpreted as structural continuity vulnerabilities rather than individual performance deficits, consistent with research identifying documentation barriers as systemic and incentive driven (Lalu et al., 2023; Vicente Sáez et al., 2020).

Interview Design

Seven fully structured qualitative interviews examined how documentation practices are enacted during project execution. Four interviews involved former student project leads from the Bakery, CRM, Cacao, and Beer bottle projects. The Beer bottle project analysed supply chain and circularity dynamics in Romania, including regulatory and infrastructural influences on resource flows. Three interviews were conducted with lab leadership, including Chris Verhoef, Rea, and Maxime Bouillon, who oversee continuity and quality across cohorts. This combination enables comparison between enacted documentation practice and institutional expectation.

A fully structured format was deliberately chosen to maintain alignment between interview prompts, validated constructs, and predefined research objectives (Kallio et al., 2016). Structured sequencing strengthens comparability and reliability while maintaining analytical rigor (DeJonckheere & Vaughn, 2019).

Construct legitimacy is supported by scholarship on student documentation practices, reuse barriers, and supervisory expectations (Chen et al., 2022; Lalu et al., 2023). Instrument design is further supported by research data management studies employing lifecycle oriented questioning and structured protocols (Kralj & Landwehr, 2025; Burnette et al., 2016; Ithaka S+R, 2019).

The interview guide was developed after theoretical validation and operationalisation. Constructs including traceability, lifecycle transparency, decision logging, validation visibility, and handover continuity were translated into fixed thematic blocks reflecting project stages. Identical question ordering reduced procedural variation and strengthened reliability, ensuring differences reflect participant experience rather than interviewer variability (Patton, 2015).

Methodological Coherence and Sufficiency

Desk research established conceptual grounding and validated the constructs used to interpret empirical observations (Peng, 2011; Wilkinson et al., 2016; OECD, 2021; Chen et al., 2022). Document analysis reveals how documentation materialises in preserved outputs. Interviews clarify enactment and perception during execution. Together, these sources form a triangulated dataset across five complete project lifecycles and seven structured interviews spanning execution and supervisory roles.

This convergence provides sufficient empirical depth to identify recurring structural patterns within the defined institutional scope. The objective is analytical transferability within applied student research environments rather than statistical generalisation. The structured and validated data collection strategy aligns with recognised standards for rigorous applied research conducted in uncontrolled real world settings (Yin, 2018; Hevner et al., 2004).

1.3 Sampling, Ethics, and Data Quality

Because this study was conducted in an uncontrolled applied research environment, credibility depends on transparent sampling, ethical safeguards, and visible quality controls rather than experimental manipulation. In qualitative case based inquiry, rigor derives from purposeful selection, construct alignment, procedural consistency, and triangulation (Patton, 2015; Yin, 2018). This section clarifies sampling logic, ethical safeguards, and data quality protection.

Sampling Strategy

Sampling was purposive and criterion based, aimed at structural insight into documentation practices within the Value Chain Hackers Lab rather than statistical representation. Purposive selection is appropriate when participants are chosen for direct relevance to the constructs under investigation (Patton, 2015).

Five completed project tracks were selected for document analysis because they represent full research lifecycles from initiation to submission. Examining complete trajectories enables assessment of documentation continuity, visibility of decision pathways, and reuse potential across entire processes rather than isolated stages, consistent with bounded case study principles (Yin, 2018).

Seven interviews were conducted: four with former student project leads from the Bakery, CRM, Cacao, and Beer bottle projects, and three with lab leadership including Chris Verhoef, Rea, and Maxime Bouillon. This dual perspective enables comparison between enacted documentation practice and institutional expectations, strengthening structural validity.

The sample size is sufficient for identifying recurring structural patterns within this bounded institutional setting. The objective is analytical insight rather than population generalisation, consistent with qualitative applied research standards (Patton, 2015; Yin, 2018).

Ethical Considerations

Ethical conduct was ensured through informed participation, transparency, and responsible data handling. All interview participants were informed of the study's purpose, academic context, and voluntary nature prior to participation, and verbal consent was obtained before recording. Findings are reported in aggregate form, and identities are not publicly disclosed beyond formal institutional roles already visible.

Document analysis relied exclusively on project materials stored within institutional repositories and educational contexts. No confidential or commercially sensitive company data is disclosed.

Data handling aligns with responsible research data management principles emphasising secure storage, controlled access, and clarity regarding reuse (OECD, 2021; Science Europe, 2021), consistent with governance expectations embedded within Dutch research infrastructure and institutional policy.

Given the researcher's familiarity with the laboratory environment, reflexive awareness of potential bias was maintained. Personal impressions were not treated as data. All analytical claims are grounded in documented project materials, recorded interviews, or converging patterns across sources, preserving separation between observation and interpretation.

Data Quality and Credibility Safeguards

Data quality was ensured through construct alignment, procedural consistency, triangulation, and source credibility.

Construct alignment was established prior to empirical interpretation. Desk research validated the constructs under investigation, including traceability, documentation completeness, decision visibility, and cohort continuity (Peng, 2011; Wilkinson et al., 2016; Chen et al., 2022). These constructs were operationalised before document analysis and interviews, reducing interpretive drift and strengthening construct validity (Gregor & Hevner, 2013).

Procedural consistency strengthened reliability. All interviews followed an identical fully structured guide with fixed ordering, directly aligned with operationalised constructs. Responses were documented immediately after interviews to reduce recall distortion, ensuring variation reflects participant experience rather than interviewer variability (DeJonckheere & Vaughn, 2019).

Triangulation was embedded at the data collection level by examining the same structural themes through document analysis, interviews with project leads, and interviews with lab leadership. Convergence across independent sources strengthens credibility, while discrepancies were treated as empirical signals rather than dismissed (Yin, 2018).

Source credibility was ensured through reliance on peer reviewed literature and recognised governance frameworks addressing reproducibility and research data management (Wilkinson

et al., 2016; OECD, 2021; Science Europe, 2021), authentic archived project outputs, and participants directly involved in the projects examined.

Sufficiency of Data

The combination of five complete project analyses and seven structured interviews provides sufficient empirical depth to identify recurring structural patterns within the defined scope of the VCH Lab. The aim is robust pattern identification within a bounded applied research environment rather than statistical saturation.

Qualitative sufficiency depends on informational richness and pattern recurrence rather than sample size alone (Patton, 2015). Across cases and participants, recurring themes concerning documentation prioritisation, continuity gaps, and incentive structures emerged, supporting the adequacy of the dataset for answering the research question within the defined institutional boundary.

1.4 Interview Evidence, Structured Summary, and Analytical Observation

Interview Design, Participants, and Scope

Fully structured qualitative interviews examined how documentation and continuity are experienced in practice within the Value Chain Hackers Lab. The full interview guide and transcripts are included in the appendix to ensure evidentiary transparency. This section synthesises and interprets the interviews while preserving traceability to original responses. All quotations are drawn directly from the transcripts.

Seven interviews were conducted. Four involved former student project leads directly responsible for research execution and final submission: Kim from the Bakery project examining sustainability practices at Bbrood; Onno from the CRM project addressing critical resource management for Gemeente Zwolle and local battery collection agencies; Quina from the Cacao project focused on transparency challenges affecting smallholder cacao farmers in African supply chains; and Kamilé from the Beer bottle project investigating supply chain and circularity considerations in Romania. These participants enacted the documentation practices under investigation and were directly involved in research decisions, analytical reasoning, material storage, and final outputs.

Three additional interviews were conducted with lab leadership, including Chris, Rea, and Maxime, who oversee research continuity and quality across cohorts. This combination enables comparison between enacted documentation practice and institutional framing, strengthening structural interpretation.

For the Bakery, CRM, and Cacao projects, accessible documentation was available for analysis alongside interviews, enabling comparison between participant descriptions and preserved

materials. The Beer bottle project did not have complete repository documentation within the study scope and therefore functions as contextual field evidence rather than a document triangulated case. This distinction clarifies evidentiary conditions and preserves transparency regarding the basis of interpretation.

Interview Findings and Observations

Across student interviews, a consistent pattern emerged: documentation was frequently positioned as supplementary to final deliverables rather than embedded in daily research activity. Kim stated that the team “mostly treated documentation as producing the final report,” that it “felt added on top,” and that “the priority was delivering the final output.” Onno observed, “You see the result, but not everything that led up to it,” explaining that when documentation “feels like an extra task instead of part of the submission process,” sustained effort declines. Quina similarly noted, “Someone can see the conclusions, but not exactly how I got there,” adding that “motivation mainly comes from assessment,” and when documentation is “not clearly graded or required,” completing the project takes precedence over recording intermediate reasoning. Kamilè reported that important shifts “were not clearly documented as a decision. It just appeared in the work later,” and that “it is not always clear why we document, so motivation is not very strong,” with students tending to “focus more on finishing the project than on documenting every step.”

Supervisory interviews reinforced this pattern at a structural level. Chris stated that “documentation takes second place” and that judges “only really value the presentation skills of the students and don’t see any of the documentation that they have produced,” adding that students “might not be motivated to document for the benefit of someone they’ll never meet” and “they’ll do the minimum.” Rea observed, “When documentation is treated as a formality, the underlying research quality is usually already compromised,” positioning documentation as integral to methodological rigor. Maxime noted that documentation “often becomes more prominent toward the end of projects” and that motivation increases when students see a “concrete purpose beyond grading.”

Across roles, a coherent structural pattern is evident. Documentation is recognised as important yet frequently treated as secondary to visible deliverables. Effort is shaped by assessment signals and clarity of purpose; where documentation is not clearly required or rewarded, intermediate analytical steps, decision rationales, and reasoning pathways are not systematically recorded.

This has direct implications for cross cohort continuity. When reasoning chains remain incomplete, subsequent teams cannot reliably reconstruct how conclusions were derived, reducing reuse potential and constraining cumulative knowledge development. These findings converge with document analysis and align with recurring patterns identified in the cross case insight analysis, where limited decision pathway visibility and inconsistent reasoning preservation were repeatedly observed.

The primary finding emerging from Section 1 is therefore triangulated and empirically grounded: documentation is not consistently embedded as an integral component of the submission process but frequently positioned as supplementary or retrospective. Within the current incentive structure, visible outputs are prioritised over systematic reasoning capture. This conclusion rests on cross role convergence, thematic recurrence across independent interviews, and alignment with preserved project materials, providing a defensible empirical foundation for the subsequent analytical phase.

Section 2. Data Analysis

2.1 Analytical Approach

2.1 Analytical Approach

The purpose of the analysis phase was to transform the collected material into defensible findings regarding documentation practices and research continuity within the Value Chain Hackers Lab. The analysis integrates interview evidence, document analysis across five completed projects, the cross case insight analysis, and desk research used to validate the constructs guiding the study. The objective was not to restate individual sources, but to determine what recurring patterns emerge across them, where evidence converges or diverges, and how these findings answer the research question.

The analysis was structured around fixed analytical dimensions established prior to interpretation. These dimensions were derived from desk research on reproducibility, research data management, and documentation quality. They include visibility of reasoning pathways, documentation completeness, decision traceability, storage coherence, and reuse potential across cohorts. These criteria reflect recognised research quality requirements rather than local preferences. Reproducibility scholarship emphasises the importance of transparent analytical provenance and traceable links between data, decisions, and conclusions (Peng, 2011). Stewardship frameworks define findability, accessibility, interoperability, and reuse as core standards for research outputs (Wilkinson et al., 2016). Lifecycle models further stress that documentation must be preserved throughout project execution to enable later interpretation and reuse (Gooch et al., 2019; Mons et al., 2017). These frameworks anchor the analysis in established research governance standards.

The analytical procedure followed three sequential steps. First, each data source was examined independently using the same fixed dimensions. Interview transcripts were reviewed systematically and coded according to themes related to documentation effort, assessment signals, decision logging, reasoning visibility, storage structure, and handover practices. In parallel, the five project cases were reviewed using identical dimensions, allowing visible documentation practices to be assessed independently from participant accounts. This step

preserved source autonomy and prevented interviews from becoming the sole basis of interpretation.

Second, cross source comparison was conducted. Student accounts were compared with supervisory perspectives to identify alignment and divergence in perceived expectations and enacted practices. Interview evidence was then compared with documented project materials and the cross case insight analysis. This process strengthens credibility by maintaining a traceable chain of evidence across multiple sources, consistent with case study quality principles (Yin, 2018).

Third, pattern testing was applied. Findings were considered structurally significant only when they recurred across multiple interviews and were supported by documentary evidence or cross case findings. Exceptions and variations were explicitly examined rather than ignored. Attention to recurrence and negative cases strengthens interpretive credibility and reduces selective bias, consistent with qualitative evaluation standards (Patton, 2015).

The study operates within an uncontrolled applied research environment and therefore does not aim for statistical measurement or frequency estimation. Instead, it follows analytic generalisation logic. Findings are grounded in evidence from a bounded institutional context and are transferable through explanation rather than statistical inference (Yin, 2018). The inclusion of projects with varied topics and partners strengthens this logic by testing whether documentation patterns persist across differing project conditions rather than being tied to a single thematic case.

Desk research functioned as an interpretive support layer during analysis. It confirmed that documentation variability, reuse barriers, and handover breakdown are recognised structural challenges in research environments (Chen et al., 2022; Lalu et al., 2023). It also provided normative standards for evaluating whether project materials preserve sufficient provenance for independent reconstruction (Peng, 2011; Wilkinson et al., 2016). Finally, it supports interpretation of documentation behaviour as shaped by structural conditions and incentive alignment rather than individual negligence (Lalu et al., 2023; Vicente Sáez et al., 2020). Desk research did not replace empirical evidence; it contextualised and disciplined interpretation.

Methodological reliability was strengthened through consistency of analytical dimensions across all sources and through the use of a fully structured interview format that ensured comparability between participants (Kallio et al., 2016; DeJonckheere & Vaughn, 2019). Interpretation was kept explicitly separate from evidence presentation. Quotations and document observations were first established, then compared across sources, and only subsequently synthesised into findings. This sequence ensures that conclusions remain visibly derived from collected material rather than assumed in advance.

The following sections apply this analytical framework to the empirical data, identify recurring structural patterns, examine divergences, and build findings that directly inform the study's conclusions and recommendations.

2.2 Analysis of Empirical Data

This section analyses the interview evidence together with the reviewed project materials to determine how documentation practices function in practice and whether recurring structural conditions are visible across projects.

Across interviews and project reviews, documentation is consistently positioned as closely tied to final submission rather than ongoing research activity. Kim, former project lead of the Bakery project at Bbrood, explained that the team “mostly treated documentation as producing the final report,” that it “felt added on top,” and that “the priority was delivering the final output.” This framing presents documentation as a delivery requirement rather than a continuous research activity. The Bakery project materials reflect this condition. While the final report is structured and readable, earlier analytical development, discarded alternatives, and intermediate reasoning steps are not consistently preserved in a form that enables full reconstruction of how conclusions were reached. The broader cross case analysis similarly identifies documentation as insufficiently detailed to support long term reuse.

A related pattern concerns the gap between outcome clarity and process transparency. Onno, who led the CRM project for Gemeente Zwolle and local battery collection agencies, stated, “A lot of ideas developed through discussions and working sessions, and those steps are not always visible in the files. You see the result, but not everything that led up to it.” His account highlights that reasoning often develops informally and is not systematically captured in preserved materials. The CRM documentation confirms this distinction. Recommendations are clearly articulated, yet the analytical development behind those recommendations, including alternative considerations and adjustments, is not consistently traceable. The cross case findings reinforce this recurring weakness in methodological transparency and documentation completeness.

Documentation of directional change presents a further limitation. Kamilè, who led the Beer bottle project examining supply chain and circularity conditions in Romania, explained, “I do not think this was clearly documented as a decision. It just appeared in the work later.” In several reviewed projects, shifts in focus are observable when comparing early drafts to final outputs. However, explicit justification for these changes is not consistently recorded. Without preserved rationale, understanding the reasoning behind strategic adjustments becomes dependent on personal explanation rather than stored evidence.

Interview evidence also indicates that documentation effort is shaped by incentive structure. Quina, who led the Cacao project focused on transparency challenges affecting smallholder farmers in African supply chains, stated, “In practice, motivation mainly comes from

assessment.” Chris, the lab supervisor, reinforced this observation, explaining that “documentation takes second place” and that judges “only really value the presentation skills of the students and don’t see any of the documentation that they have produced.” The alignment between student and supervisory accounts suggests that documentation effort responds to what is visibly rewarded during evaluation. This dynamic is reflected in the project materials. Final submissions are polished and structured, whereas earlier reasoning steps are less consistently preserved.

Maxime further observed that documentation “often becomes more prominent toward the end of projects.” The reviewed materials support this timing effect. Structured reporting intensifies near submission, while earlier analytical stages show less systematic preservation. Documentation therefore appears to function retrospectively rather than as an embedded, ongoing practice.

Rea explicitly linked documentation to research quality, stating, “When documentation is treated as a formality, the underlying research quality is usually already compromised.” This concern is supported by the project reviews. Although conclusions are presented clearly, the limited visibility of reasoning pathways reduces the ability of an independent reviewer to verify how evidence supports those conclusions.

Taken together, the convergence of student perspectives, supervisory accounts, and reviewed project materials reveals a coherent structural pattern. Documentation is largely positioned as part of final reporting rather than integrated into daily research work. Conclusions are visible, yet the reasoning steps that produced them are not consistently preserved. Directional shifts occur but are not always logged with explicit justification. Documentation effort responds to grading signals and intensifies near submission rather than being maintained throughout the project lifecycle. The recurrence of these conditions across multiple interviews and project cases indicates that this represents a structural feature of the current research environment rather than isolated behaviour by individual teams.

2.3 Identified Patterns and Insights

The analysis in Section 2.2 demonstrates that the observed documentation weaknesses are not isolated incidents but form a recurring structural pattern shaped by workflow structure, incentive alignment, and evaluation visibility. This section interprets those findings and situates them within recognised research quality frameworks.

The first structural pattern is the separation between research activity and documentation activity. Documentation is concentrated around final submission rather than embedded throughout project execution. Analytical reasoning develops through discussion, iteration, and adjustment, yet documentation primarily captures stabilised outcomes. This separation limits transparency and reduces the traceability of how conclusions were formed. Reproducibility

scholarship emphasises that credible research depends on preserved analytical pathways, not only clear results (Peng, 2011). The FAIR principles further stress that reuse depends on accessible and interpretable documentation of reasoning processes (Wilkinson et al., 2016). When intermediate reasoning is not preserved, reuse capacity declines even if final outputs appear structured.

The second pattern concerns the absence of systematic decision logging. Directional changes occur during projects, but explicit justification for these changes is not consistently recorded. Lifecycle models of research data management emphasise that documentation should preserve contextual reasoning alongside outputs to maintain interpretability across time (Gooch et al., 2019; Mons et al., 2017). Without recorded decision rationale, accountability weakens and continuity becomes dependent on personal explanation rather than stored evidence.

The third pattern relates to incentive structure. Documentation effort is shaped by grading signals and evaluation visibility. When presentation quality is more visible during judging than documentation depth, effort rationally shifts toward presentation. Research on research practice barriers shows that documentation behaviour is strongly influenced by structural incentives (Lalu et al., 2023). In this environment, documentation is recognised as important but is not consistently embedded as a weighted and ongoing requirement, which explains its secondary positioning in practice.

The fourth pattern concerns timing. Documentation intensifies near submission rather than being maintained throughout project execution. This creates a retrospective reconstruction effect in which reasoning is organised at the end to produce a coherent narrative rather than preserved continuously as it develops. Reproducibility frameworks consistently emphasise contemporaneous documentation rather than post hoc reconstruction (OECD, 2021; Science Europe, 2021). Retrospective documentation improves readability but weakens evidentiary authenticity.

The fifth pattern concerns continuity risk. Because reasoning steps and decision justifications are not consistently preserved, future teams rely on informal handover or personal memory. Research on knowledge transfer and research data stewardship highlights that sustainable research systems depend on preserved contextual information rather than informal explanation (Burnette et al., 2016; Vicente Sáez et al., 2020). Where contextual reasoning is incomplete, cumulative knowledge development is constrained.

Across interviews and project cases, no participant described documentation as fully embedded, continuously recorded, and structurally integrated from start to finish. Although emphasis varied, the absence of contradiction strengthens credibility. The convergence of student and supervisory perspectives indicates that the condition is systemic rather than behavioural.

Taken together, the evidence supports a central structural insight: documentation practices within the current environment are shaped less by lack of awareness and more by workflow design and incentive alignment. Students recognise documentation, supervisors value it, yet it is

not consistently embedded as an ongoing research activity. As a result, reasoning visibility, decision traceability, and reuse capacity remain structurally limited. Addressing this condition therefore requires structural intervention rather than individual correction.

2.4 Limitations and Scope Clarification

While the analysis reveals consistent structural patterns, several boundaries define the scope and interpretation of the findings.

First, the empirical data consists of five completed project cases and seven interviews conducted within the Value Chain Hackers Lab. The findings therefore describe structural conditions within this specific applied research environment. The study does not claim statistical generalisation beyond this context. Instead, conclusions are analytically transferable to comparable student based applied research settings.

Second, document analysis relied on preserved materials available at the time of review. In the Beer bottle project, complete repository documentation was not accessible, and findings for that case rely primarily on interview evidence. However, similar documentation patterns were confirmed across the other reviewed projects, reducing the likelihood that this single limitation materially alters the overall interpretation.

Third, interviews reflect participant perspectives shaped by retrospective reflection. To strengthen credibility, interview statements were compared with preserved project materials wherever possible. Convergence between reported practice and stored documentation increases confidence in the reliability of the findings.

Fourth, the researcher's familiarity with the laboratory environment required reflexive awareness of potential bias. Interpretations were grounded strictly in interview transcripts, documented materials, and cross case comparison. Patterns were identified only when supported by multiple sources.

The interpretation of the identified condition as structural is further supported by institutional and system level benchmarking. At national level, funding bodies such as NWO require formal research data management planning at project inception (NWO, n.d.). At institutional level, universities such as the University of Twente align research practice with FAIR principles and defined documentation responsibilities (University of Twente, n.d.). At infrastructure level, SURF supported platforms such as Yoda require structured metadata prior to archival preservation (SURF, n.d.). Operational verification services at TU Delft assess reproducibility of research processes (TU Delft Digital Competence Centre, n.d.), while long term infrastructures such as 4TU.ResearchData provide structured publication and persistent identifiers (4TU.ResearchData, n.d.). Workflow initiatives such as The Alan Turing Institute's Turing Way embed documentation into daily research routines (The Alan Turing Institute, n.d.).

Across these levels, documentation is structurally embedded within governance, infrastructure, and verification mechanisms. This benchmarking does not extend the empirical sample but provides contextual validation that the observed condition reflects a documentation maturity gap rather than isolated team behaviour.

Finally, this study evaluates documentation visibility and reasoning traceability, not the substantive correctness of project conclusions. It assesses whether reasoning pathways are sufficiently preserved to allow independent understanding and reuse, rather than whether recommendations were technically right or wrong.

These boundaries define the scope without weakening the findings. The convergence of interviews, project reviews, cross case comparison, and benchmarking supports the credibility of the conclusions while clarifying what the study does and does not claim.

Understood. Below is a fully concise version of **Section 3.1**. All research questions are answered clearly. Logical progression is preserved. Citations are used sparingly and not repeated unnecessarily. Redundancy is removed, but justification remains intact.

Section 3: Conclusions and Recommendations

A fully elaborated version of Section 3, including the complete analytical breakdown, extended theoretical justification, detailed benchmarking logic, and expanded framework specification, is provided in Appendix E. The present section presents a structured and concise synthesis of those findings while preserving the full logical chain from diagnosis to validated framework design.

3.1 Answers to the Research Questions

Main Research Question

How can the Value Chain Hackers Lab design and validate a reproducible research framework that makes student projects transparent, verifiable, and reusable in a consistent and institutionally reliable way?

Section 2 demonstrated that current projects do not consistently achieve transparency, verifiability, or reuse. Documentation is often concentrated in final reporting. Decision rationales are not systematically logged. Intermediate reasoning steps are difficult to reconstruct. As a result, reuse across projects is limited.

Reproducibility scholarship shows that credible research depends on visible analytical pathways and preserved provenance rather than readable outcomes alone (Peng, 2011; Wilkinson et al., 2016). Governance frameworks similarly embed documentation as a structural requirement rather than optional practice (OECD, 2021).

The Lab must therefore implement a structured framework that embeds documentation throughout the research lifecycle, standardises file organisation and metadata, introduces mandatory decision logging and supervisor checkpoints, and aligns documentation quality with assessment criteria. Validation must be based on whether an independent reviewer can reconstruct reasoning pathways and whether documentation meets core reuse standards. This directly addresses the diagnosed structural weaknesses.

Subquestion 1 — Define

What structural and procedural barriers have limited reproducibility in past and current VCH projects?

Document analysis and structured interviews revealed recurring structural barriers. Documentation is not consistently embedded in workflow. Decision reasoning is not always logged. Links between sources, analysis, and conclusions are not always visible. File structures vary across projects. Documentation effort intensifies near submission rather than remaining stable during execution.

These findings indicate systemic misalignment rather than individual failure. Work System Theory explains that outcomes emerge from the interaction of roles, tools, incentives, and processes (Alter, 2013). Studies on student documentation confirm that incomplete provenance and inconsistent data management typically stem from structural design gaps rather than negligence (Chen et al., 2022). Reproducibility breakdown is therefore embedded in workflow and incentive structure.

Subquestion 2 — Design

Which reproducibility practices from other research environments can inform an approach suitable for the VCH Lab?

Benchmarking showed that mature research environments embed documentation early in project design, require structured metadata, define stewardship roles, and integrate review mechanisms. FAIR positions reuse as a core quality standard, while national governance bodies require formal data management planning from project inception (Wilkinson et al., 2016; OECD, 2021).

The transferable principle is clear. Reproducibility maturity emerges from governance structure, not individual discipline. Documentation becomes reliable when it is embedded, checked, and institutionally reinforced.

Subquestion 3 — Design

How can these principles be translated into a practical reproducible research framework for students and supervisors?

Translation requires converting governance logic into operational structure suitable for applied student research. Design Science Research requires solutions to be grounded in diagnosed problems and evaluated against explicit criteria (Hevner et al., 2004).

The framework must define lifecycle stages, introduce structured decision logs, standardise file naming and metadata, embed supervisor verification checkpoints, and align grading with documentation quality. Lifecycle models emphasise that documentation must span planning, execution, analysis, and reporting rather than appear only at final submission (Mons et al., 2017). The stage reference model supports distributing documentation duties across phases rather than concentrating them at the end.

The framework is therefore grounded both in empirical diagnosis and established research governance principles.

Subquestion 4 — Execute

How can the framework be tested for clarity, feasibility, and alignment with institutional policies?

Testing requires measurable evaluation logic. Validation must assess whether independent reviewers can reconstruct reasoning pathways, whether documentation completeness improves compared to prior projects, and whether practices align with recognised governance standards.

FAIR provides reuse and accessibility criteria. Governance frameworks provide compliance benchmarks (Wilkinson et al., 2016). Evaluation must therefore be structured and criteria based rather than subjective.

Subquestion 5 — Learn

How can the framework support long term knowledge continuity and educational value within the Lab?

Continuity requires documentation practices to become routine rather than temporary. Communities of Practice theory explains that stable practices emerge through shared norms and reinforcement (Wenger, 1998). Knowledge creation theory emphasises structured preservation and transfer for institutional learning (Nonaka and Takeuchi, 1995).

Because current documentation does not reliably support independent reconstruction or cumulative knowledge, long term value depends on structured repositories, defined responsibilities, and incentive alignment. Sustainability is therefore both structural and social.

Research Progression During GP2

During GP2, additional analytical questions deepened the diagnosis.

First, structural breakdown was traced to interacting elements including workflow design, unclear role ownership, limited validation checkpoints, and weak incentive alignment. This reframed documentation weakness as system interaction rather than isolated habit.

Second, stage based analysis clarified where breakdown occurs. Documentation appears strongest at final reporting and weaker during earlier execution stages. Lifecycle thinking enabled identification of stage specific vulnerability rather than treating documentation as binary.

Third, institutional benchmarking confirmed that the Lab condition reflects documentation maturity gaps rather than local anomaly. Mature environments embed documentation through planning requirements, metadata standards, and verification structures.

Fourth, framework design requirements were derived directly from this diagnosis. Mandatory decision logging, lifecycle structuring, supervisor checkpoints, metadata standards, and assessment alignment are not arbitrary features but structural corrections.

Fifth, long term embedding was reframed as organisational normalisation rather than template introduction. Sustainable change requires collective reinforcement and routine integration.

3.2 Synthesis and Strength of Conclusions

The central conclusion of this research is that the Value Chain Hackers Lab does not yet operate with a structured system that consistently guarantees transparency, verifiability, and reuse across student research projects. While final reports are typically readable and complete,

the reasoning pathway from sources to analysis to conclusions is not consistently preserved in a way that allows independent reconstruction.

This conclusion rests on convergence across structured document analysis, structured interviews, and institutional benchmarking. Across five completed projects, final deliverables were present, yet intermediate reasoning steps and decision rationales were not consistently recorded as explicit and traceable elements. Analytical development could often be inferred, but not independently verified. A new student or supervisor could see what was concluded, but not reliably how those conclusions were reached.

Interview evidence reinforced this pattern. Students described reasoning processes that occurred but were not consistently captured in stored documentation. Supervisors confirmed that documentation frequently takes second place to presentation and visible performance in evaluation settings. Documentation effort was reported to increase near submission rather than being embedded throughout execution. The alignment between execution level and supervisory level accounts strengthens internal validity.

The pattern is structurally explainable. Work System Theory demonstrates that behaviour follows the alignment of roles, tools, processes, and evaluation signals. If assessment visibly rewards final output over documented reasoning, students will rationally prioritise presentation elements. Socio technical systems theory further clarifies that durable practice requires both structural tools and social reinforcement. Without enforced checkpoints and embedded expectations, documentation remains discretionary rather than routine.

These findings align with established reproducibility standards. FAIR defines reuse and transparency as core dimensions of research quality. Reproducible research requires visible analytical pathways rather than merely readable outcomes. Lifecycle models show that documentation must span planning, execution, analysis, and reporting stages instead of appearing retrospectively. When stage awareness is applied, documentation within the Lab appears strongest at final reporting and weaker during earlier working stages.

Institutional benchmarking confirms that this condition is structural rather than isolated. National funders and research governance bodies embed documentation through early planning requirements, metadata standards, and structured review mechanisms. Compared to these mature environments, the Lab currently relies more on student discretion than on enforced structure.

The strength of the conclusion lies in convergence. Project reviews, student interviews, supervisor interviews, theoretical explanation, and governance benchmarking all point toward the same structural misalignment. The study remains appropriately bounded to one applied research lab, yet alignment with recognised reproducibility standards makes the findings analytically transferable to similar student research environments.

The core issue is therefore structural embedding. Documentation is not yet integrated as a visible, guided, and assessed component of research workflow. Without such embedding,

transparency remains partial, verifiability limited, and reuse unreliable. This directly necessitates the structured framework specified in Section 3.3.

3.3 Specifications and Criteria for Solution Development

The proposed solution is a reproducible research framework integrated as Lab level policy. A policy approach is necessary because documentation behaviour is shaped by what is expected, monitored, and rewarded. When documentation is treated as secondary, students logically prioritise visible outputs. Structural embedding replaces discretion with consistency and aligns behaviour with institutional goals.

The framework follows design science logic. Diagnosed breakdowns are translated into explicit structural requirements evaluated against defined success criteria. The framework aligns with FAIR and broader research governance standards that treat documentation as a core research responsibility. A lifecycle lens ensures documentation occurs across project stages rather than only at final reporting. Stage awareness is used solely to verify that documentation is captured during identification, processing, review, and production phases. Provenance principles reinforce that decision history and source use must remain visible if work is to be verified and reused.

The policy intent is to guarantee transparency, verifiability, reuse, and consistency across all Lab projects. Transparency means reasoning pathways and source use are visible. Verifiability means conclusions can be independently reconstructed. Reuse means a new team can continue work using preserved materials. Consistency means the same enforced structure applies to every project.

The framework begins with structured onboarding. Each project starts with a standardised workspace including defined folder structure, file naming conventions, and required decision and source logs. Early structural setup prevents later reconstruction gaps and aligns with lifecycle documentation principles. The student project lead establishes the workspace, and the supervisor verifies compliance before research proceeds.

During execution, documentation is maintained continuously rather than retrospectively. Students record key decisions, shifts in direction, and how sources inform claims through structured but concise log entries. The goal is not volume but visibility of reasoning. The project lead ensures completeness, and supervisors review progress during scheduled checkpoints.

Supervisor checkpoints occur before final submission to verify traceability. These checks confirm that decisions are recorded, sources are linked to claims, and analytical steps are understandable without verbal explanation. Early verification shifts documentation from end stage correction to embedded practice.

Final submission integrates documentation into the deliverable. The project is not considered complete unless it includes both the final report and the structured documentation package, including decision log, source log, and organised research materials. Documentation becomes assessed output rather than optional addition.

Before archiving, projects are prepared for reuse through structured metadata that clarifies project aim, methods, key sources, and log locations. This ensures long term findability and interpretability and aligns with repository and governance expectations.

The framework succeeds only if transparency, verifiability, reuse, and institutional consistency can be independently assessed. An independent reviewer must be able to follow the reasoning pathway without private explanation. A new team must be able to continue work using archived materials. The structure must apply uniformly across projects so that documentation quality is not dependent on individual habit.

By embedding documentation into workflow, checkpoints, and evaluation logic, the framework directly addresses the structural weaknesses identified in Section 2. It is enforceable, measurable, and aligned with recognised governance standards.

3.4 Validity, Certainty, and Remaining Unknowns

The conclusions are supported by converging evidence from structured case analysis, structured interviews, and staged desk research aligned with recognised governance and reproducibility standards. Across five diverse projects, similar weaknesses in reasoning visibility, decision logging, and storage structure were observed. Seven interviews across student and supervisory roles described the same structural condition. Alignment with established research on documentation breakdown and incentive driven behaviour strengthens external validity.

The framework is not speculative. It is logically derived from diagnosed breakdowns, grounded in system and design science logic, and benchmarked against national and institutional documentation standards. Its requirements are traceable to empirical findings and governance expectations.

Certainty is strong within the defined institutional scope. The study does not claim universal generalisation but provides analytically grounded conclusions about one applied research lab. Convergence across evidence layers reduces the likelihood of misinterpretation.

Remaining uncertainty concerns implementation rather than diagnosis. Long term behavioural embedding must be observed in practice, as stable routines require reinforcement. Full integration into broader university systems may require coordination beyond this study. Although the framework integrates documentation into evaluation logic, behavioural change must be assessed after enforcement begins.

Within scope, the diagnosis of structural documentation weakness is highly certain. The framework design is logically derived and externally validated. Remaining unknowns concern long term adoption and institutional embedding, not the validity of the analysis or the justification of the solution.

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